

Statistic  
framework of  
the flexible  
RNA

Wandrille  
Buchy

Goals of the  
validation

State-of-the-  
Art

Dimensionality  
reduction  
 $\eta$  -  $\theta$  pseudo-angles

Results

$\eta$  -  $\theta$  limitations

Goal of the  
dependency

Goal  
Exemple of a  
representation

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Next steps

# Dimensionality reduction methods on the flexible RNA molecules

Wandrille Buchy

ENSAI, LAAS

18.06.2026



# Goals

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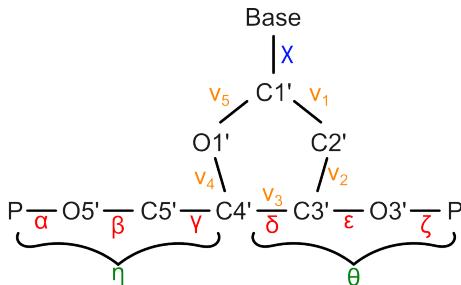
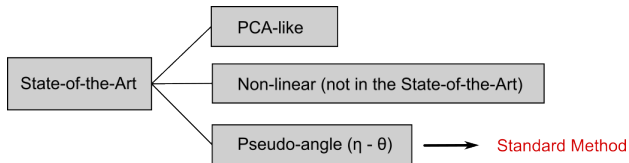
Next steps

## Goal

$(\alpha_1, \dots, \alpha_K) \in \mathbb{T}^K, 7 \leq K \leq 12 \rightarrow$  Two-dimensional Space

- **Goal 1** : Evaluation of the State-of-the-Art.
- **Goal 2** : Study of the dependency between the angles to build another representation.

# Dimensionality reduction



# $\eta - \theta$ pseudo-angles

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> J Mol Biol. 2007 Sep 28;372(4):942-957. doi: 10.1016/j.jmb.2007.06.058. Epub 2007 Jun 27.

## Evaluating and learning from RNA pseudotorsional space: quantitative validation of a reduced representation for RNA structure

Leven M Wadley <sup>1</sup>, Kevin S Keating <sup>2</sup>, Carlos M Duarte <sup>3</sup>, Anna Marie Pyle <sup>4</sup>

► RNA. 2023 Dec;29(12):1896–1909. doi: [10.1261/rna.079821.123](https://doi.org/10.1261/rna.079821.123) 

### The pseudotorsional space of RNA

[Leandro Grille](#) <sup>1,2,5</sup>, [Diego Gallego](#) <sup>3,4,5</sup>, [Leonardo Darré](#) <sup>2,3</sup>, [Gabriela da Rosa](#) <sup>1,2</sup>, [Federica Battistini](#) <sup>3,4</sup>, [Modesto Orozco](#) <sup>3,4,</sup>, [Pablo D Dans](#) <sup>1,2,3,</sup>

→ We tried to do better !

# Validation results

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|                    | Cluster<br>preservation | Distance<br>preservation | Local<br>neighbourhood<br>preservation | Global<br>neighbourhood<br>preservation |
|--------------------|-------------------------|--------------------------|----------------------------------------|-----------------------------------------|
| GeoPCA             | X                       | X                        | X                                      | ✓                                       |
| t-SNE              | ✓                       | X                        | X                                      | ✓                                       |
| UMAP               | ✓                       | ✓                        | X                                      | ✓                                       |
| $(\eta - \theta)$  | X                       | X                        | X                                      | ✓                                       |
| $(\alpha - \zeta)$ | X                       | ✓                        | X                                      | ✓                                       |



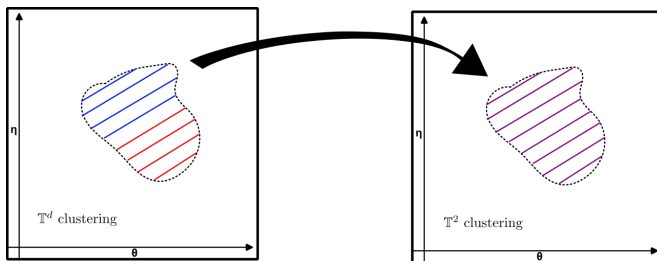
Mostly positive



Mostly negative

# $\eta - \theta$ limitations

⚠ 2 clusters in  $\mathbb{T}^d \rightarrow 1$  cluster in  $\mathbb{T}^2$ .



## Conclusion

We lose information so we need a different method.

*Additional analysis confirm such phenomenon and its independence with the conformation.*

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# Goal

## Goal

For each pair of angles  $(\alpha, \beta)$ , we want to test :

$$H_0 : \alpha \perp \beta$$

**Output**  
**(Per angle**  
**pair)**

**"Distance to independence"**

**p-value for  $H_0 : \alpha \perp \beta$**

## Remark

The classical test methods are not usable for  $\mathbb{S}^1$  so we used an Optimal Transport based method.

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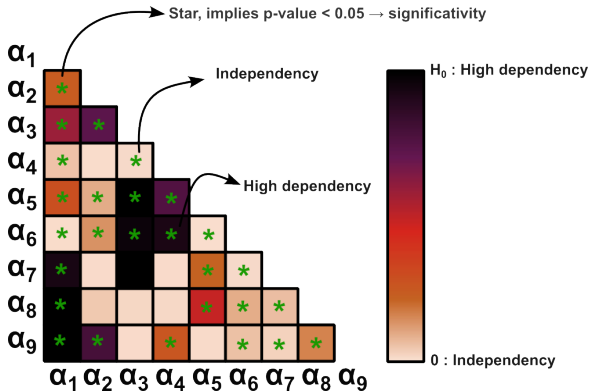
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# Exemple of a representation



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# Results of the tests

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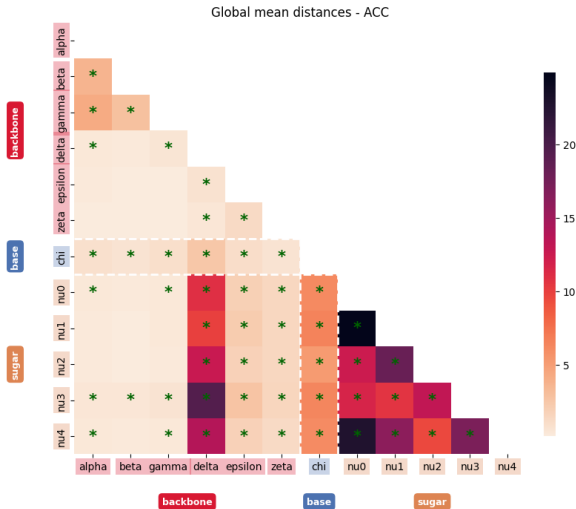
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# Next steps

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Next steps

- Study of the independence between groups of angles.
- Proposition of a new pseudo-angle approach with the dependency found.